Cosmic Radiation Lab Course

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1 Preparation

1.1 Cosmic Radiation

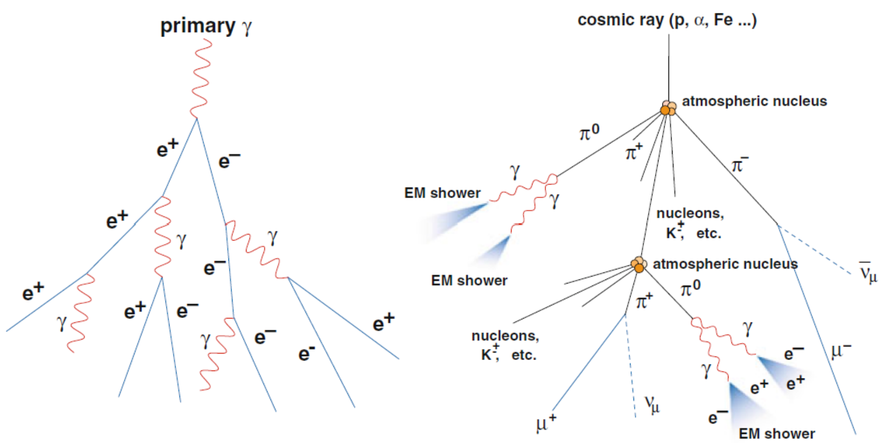

Figure 1: Schematic description shower of atmospheric right hadronic and left electromagnetic shower [2]

Physicist Victor Hess made balloon flight (5300 metres) that opened new window on matter in universe. He measured rate of ionisation in atmosphere & found that it increased to some three times that at sea level. He concluded radiation entering atmosphere from atmosphere then discovered cosmic rays. Cosmic rays are high-energy particles that come from space [1]. The majority of high-energy particles in cosmic rays are protons (H nuclei) about 89% , Helium nuclei (alpha particles) 10%, & 1% Heavier nuclei (up to uranium). When cosmic rays arrive at Earth, they collide with nuclei of atoms in upper atmosphere, creating more particles, mainly pions. The charged pions decay, emitting particles muons [4]. These together account for 99% of cosmic rays, and electrons, photons, and remaining 1% neutrinos [2]. Cosmic rays touching atmosphere (primary cosmic rays) produce secondary particles that can arrive Earth surface, through multiplicative showers. But most cosmic ray primaries are influenced by solar magnetic field, most of those detected near Earth have kinetic energies in excess of about 0.1 GeV. At 10^{15} eV, GCRs consist mostly of protons (nuclei of H atoms) and alpha particles (He nuclei). UHECRs consist mostly of charged nuclei. Gamma rays observed with energies as high as $\sim 10^{12}$ eV. EAS generated by photons would be almost purely electromagnetic. There are two types of cosmic rays primary and secondary radiation. **Electromagnetic** & **Hadronic showers** described in Figure 1 Electromagnetic showers can produced by primary γ ray. Photons lose energy in the atmosphere by pair production, produce more photons by Bremsstrahlung radiation. Hadronic showers produced by primary charged particle cosmic ray. This creates pions & kaons that then decay. The distribution energy (spectrum) of cosmic rays explained by power law E^{-p}, with so-called spectral index p around 3 on average. Energy up to about 10^{21} eV (GZK cutoff) [14].

1.2 Muons Production and detection

In this experiment our aim is to detect secondary cosmic ray particles muons. Charged pions decay channel to muons process which are secondary particles of cosmic rays

$$\pi^+ \rightarrow \mu^+ + \nu_\mu \quad \& \quad \pi^- \rightarrow \mu^- + \overline{\nu}_\mu \tag{1}$$

Charged pion $\pi^\pm$ decays into muon corresponding muon neutrino ν_μ and $\overline{\nu}_\mu$

Charged kaon second channel produce muons.

$$K^+ \rightarrow \mu^+ + \nu_\mu \quad \& \quad K^- \rightarrow \mu^- + \overline{\nu}_\mu \tag{2}$$

Due to their relatively long lifetime ($\tau_\mu \approx 2.2\,\mu$s) & low interaction cross-section with matter, muons can penetrate to sea level & detected by scintillator.

We will use scintillator for detect muons.

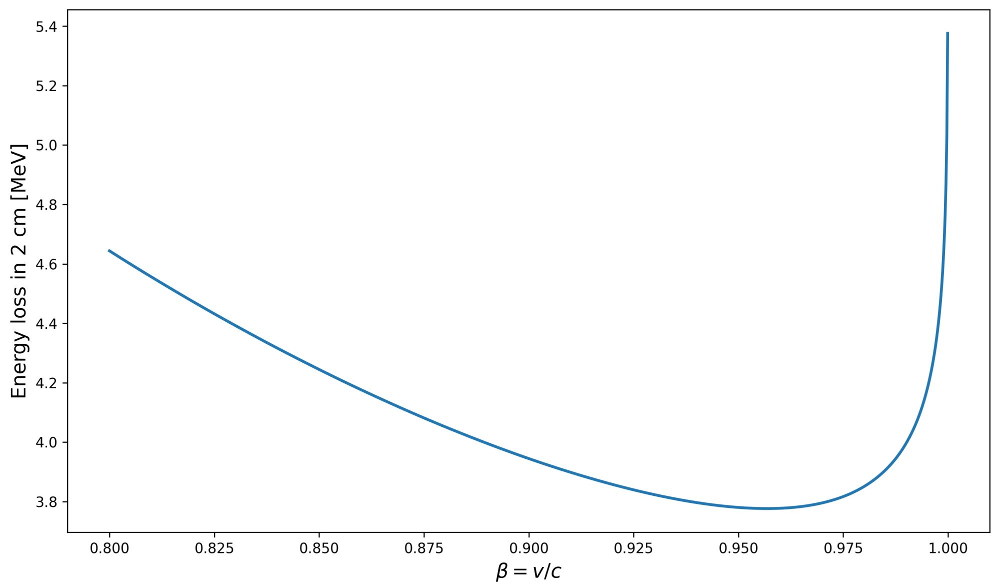

Figure 2: Energy loss of muons in 2 cm of scintillator vs $\beta = v/c$.

Energy loss of muons in 2 cm of scintillator (use Bethe-Bloch-Formula assuming scintillator consists of hydrogen and carbon atoms with $\rho = 1.032\,\mathbf{g/cm}^3$

The mean energy loss per unit path length for muons in a medium is given by Bethe-Bloch formula [10]

$$-\frac{dE}{dx} = \frac{z^2 e^4 n_e}{4\pi\varepsilon_0^2 m_e v^2} \cdot \left(\ln\left(\frac{2m_e v^2}{E_B^{(e)}(1-\beta^2)} \right) - \beta^2 \right) \tag{3}$$

where:

- z: charge number of the incident particle
- e: elementary charge
- $n_e = n_v Z = \frac{Z}{A}\rho N_A$: electron density where Z atomic number, A atomic mass & ρ mass density
- ε_0: vacuum permittivity
- m_e: electron mass

- v: velocity of incident particle
- $\beta = v/c$: relativistic velocity
- $E_B^{(e)}$: mean excitation energy of material

where $\rho = 1.032\,\mathrm{g/cm^3}$, $E_B^{(e)} \approx 13.5\,\mathrm{eV} \times Z_{\mathrm{abs}}$

Now we can calculate Energy loss of muons in 2 cm of scintillator where Z_{abs} is atomic number of absorber.

For carbon: $\frac{Z}{A} = 6/12 = 0.5$
Atomic Number for Carbon $Z = 6$
Mean Excitation Energy
$E_B = 13.5$ eV $\cdot Z = 13.5$ eV $\times 6 = 81$ eV
Muon velocity for minimal energy loss 3.78Mev is $\beta = 0.9569$, which corresponds to kinetic energy for muon :

$$E_{\mathrm{kin}} = \gamma m_\mu c^2 - m_\mu c^2 = 258\,\mathrm{MeV} \tag{4}$$

1.3 Radioactive Decays

In experiment detector will calibrated β-decay of Cobalt-60 (^{60}Co) & Sodium-22 (^{22}Na). β-decay comes in two forms: β^+-decay and β^--decay. β^+ decay

$$^A_Z\mathrm{X} \to {}^A_{Z-1}\mathrm{X}' + e^+ + \nu_e$$

β^- decay

$$^A_Z\mathrm{X} \to {}^A_{Z+1}\mathrm{X}' + e^- + \bar{\nu}_e$$

After the β-decay the nucleon de-exites and emits a γ-ray that can then be detected in the scintillator by for example the Compton effect.

Calculate Compton edge using mean energy from both decays

The energy transferred to an electron in Compton scattering is given by:

$$E_e = \frac{E_\gamma \cdot \varepsilon(1 - \cos\vartheta)}{1 + \varepsilon(1 - \cos\vartheta)} \tag{5}$$

where:

- E_γ is the energy of the incident photon
- ϑ is the scattering angle of the photon
- $\varepsilon = \dfrac{E_\gamma}{m_e c^2}$ with $m_e c^2 = 0.511\,\mathrm{MeV}$

Maximum Energy Transfer (Compton Edge)
The maximum energy transfer occurs in a central collision where the photon is back scattered:

$$\vartheta = 180^\circ \quad \Rightarrow \quad \cos\vartheta = -1$$

Substituting into equation (5):

$$E_e^{\max} = \frac{E_\gamma \cdot \varepsilon(1-(-1))}{1+\varepsilon(1-(-1))}$$

$$= \frac{E_\gamma \cdot \varepsilon(2)}{1+\varepsilon(2)}$$

$$\boxed{E_e^{\max} = \frac{2\varepsilon \cdot E_\gamma}{1+2\varepsilon}}$$

This maximum energy transfer corresponds to **Compton edge** observed in gamma-ray spectra, which is used for energy calibration in this experiment.

Now calculate Compton Edge for ^{60}Co and ^{22}Na atoms-

For ^{22}Na

β^+-decay of ^{22}Na have γ photon with energy 1.2745 MeV:

$$\varepsilon = \frac{1.2745}{0.511} = 2.495$$

$$E_{e,^{22}\mathrm{Na}} = \frac{2 \times 2.495 \times 1.2745}{1 + 2 \times 2.495}$$

$$= \frac{6.357}{5.990} = 1.06\,\mathrm{MeV} \tag{6}$$

For ^{60}Co have two gamma quanta with energies 1.173 MeV & 1.332 MeV Energy. They are first averaged since they are not resolved by the detector, following the lab manual [5].

$$E_\gamma^{\text{mean}} = \frac{1.1732 + 1.3325}{2} = 1.25285\,\mathrm{MeV}$$

$$\varepsilon = \frac{E_\gamma^{\text{mean}}}{m_e c^2} = \frac{1.25285}{0.511} = 2.452$$

$$E_{e,^{60}\mathrm{Co}} = \frac{2 \times 2.452 \times 1.25285}{1 + 2 \times 2.452}$$

$$= \frac{6.143}{5.904} = 1.04\,\mathrm{MeV} \tag{7}$$

[11, 12]

1.3.1 Landau Distribution

It explains charged particles energy loss during pass through matter.

Landau Distribution function, described by :

$$p(x) = \frac{1}{2\pi i} \int_{a-i\infty}^{a+i\infty} e^{s \ln s + xs} \, ds, \tag{8}$$

where a = arbitrary positive real number that means integration path parallel to imaginary axis, & ln is natural logarithm.

The real integral :

$$p(x) = \frac{1}{\pi} \int_0^{\infty} e^{-t \ln t - xt} \sin(\pi t) \, dt \tag{9}$$

When ionizing particle moves through thin detector medium, fluctuations explained by Landau distribution is probability distribution.[3] The derivation of distribution according to Landau is based on consideration that a fast particle traverses a layer of matter.
Distribution of energy losses by ultrarelativistic charged particles due to ionization of the media.[7]

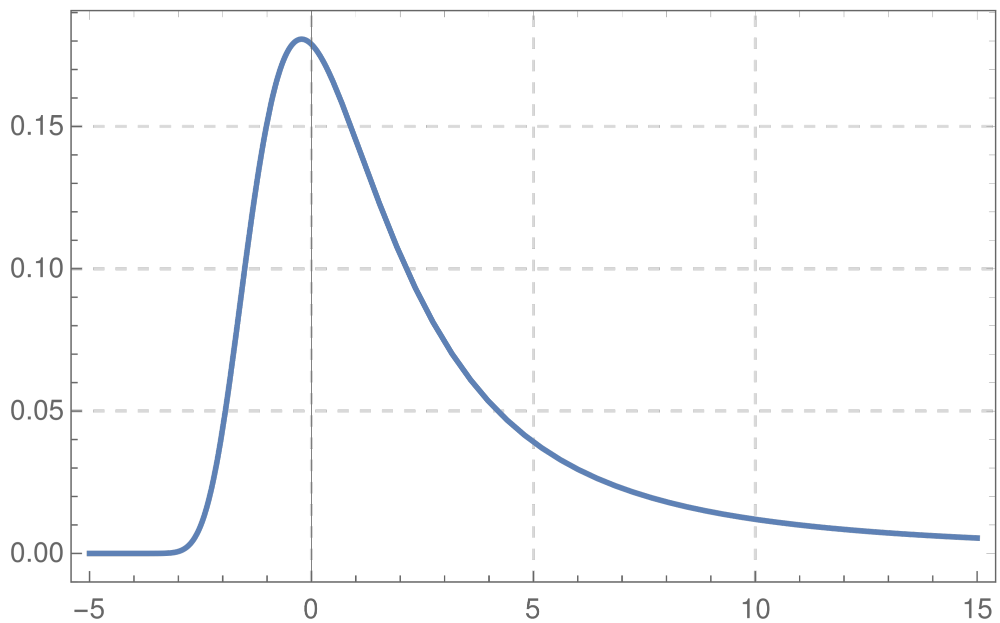

Figure 3: Probability density function of the Landau distribution. The data points show the characteristic asymmetric peak and long tail [3].

1.4 Light Matter Interaction

High-energy photons interact with matter via photoelectric, Compton scattering, & electron–positron pair production [10].

Photoelectric Effect

The photoelectric effect is ejection of an electron from a material that has just absorbed a photon. The ejected electron is called a photoelectron.
Absorption of gamma quantum with complete energy transfer to an atomic electron

Emission of electron with

$E_e = E_\gamma - E_B^{(i)} \quad i = K, L, M \ldots$

where: E_γ is energy of incident gamma photon, $E_B^{(i)}$ is binding energy of electron in shell i (K, L, M, etc.) & E_e is kinetic energy of emitted photoelectron.

At low energies: Absorption edges

Edges position follow Moseley's law:

$$E_B^{(K)} = k(Z - s_K)^2$$

where s_K is screening constant.

The cross section decreases with increasing gamma energy

Most important for inner shells (K-shell: $\approx 80\%$)

The nucleus must absorb excess momentum
Strong Z-dependence
Cross Section

$$\mu_{\text{Photo}} \propto \frac{Z^5}{E_\gamma^{7/2}}$$

The photoelectric absorption coefficient exhibits dependence on atomic number Z & photon energy E_γ

Compton scattering

Elastic scattering between photon & bound electron, resulting in change in direction & energy of both gamma quantum & electron.

The interaction can be simplified under condition:

$$E_\gamma \gg E_B^{(i)}$$

where: E_γ is incident photon energy

$E_B^{(i)}$ is binding energy of e^- electron

From energy & momentum conservation, following relation for scattered photon energy follows:

$$E'_\gamma = \frac{E_\gamma}{1 + \varepsilon(1 - \cos\vartheta)} \tag{10}$$

with

$$\varepsilon = \frac{E_\gamma}{m_e c^2}$$

where:
E_γ is energy of incident photon, E'_γ is energy of scattered photon & ϑ is scattering angle of photon.

The energy transferred to electron in Compton scattering is given by:

$$E_e = \frac{E_\gamma \cdot \varepsilon(1 - \cos\vartheta)}{1 + \varepsilon(1 - \cos\vartheta)} \tag{11}$$

Maximum energy transfer occurs in a central collision where the photon is back scattered:

$$\vartheta = 180^\circ \quad \Rightarrow \quad \cos\vartheta = -1$$

where:

E_γ is the incident photon energy, $\varepsilon = \dfrac{E_\gamma}{m_e c^2}$ with $m_e c^2 = 0.511\,\text{MeV}$ & ϑ is the scattering angle of the photon

Substituting into equation (11):

$$E_e^{\max} = \frac{E_\gamma \cdot \varepsilon(1-(-1))}{1+\varepsilon(1-(-1))}$$

$$\boxed{E_e^{\max} = \frac{2\varepsilon \cdot E_\gamma}{1+2\varepsilon}}$$

The differential cross section $d\sigma_{\text{Compton}}$ described as Photon energy , Scattering angle, & Polarization direction.

A key dependence of Compton attenuation coefficient is:

$$\mu_{\text{Compton}} \propto \frac{Z}{E_\gamma}$$

The Compton effect Dominates up to approximately 10 MeV. Gains significance compared to photoelectric effect with increasing gamma energy

Pair production

Near nucleus photon creating an $e^+ - e^-$ pair production refers. Interaction of incoming energy must be above threshold of at least total rest mass energy of two particles, and conserve momentum & energy both. In photon–matter interactions with photon energy probability of pair production increases & also nearby atom increases approximately square of atomic number. [8]

Emission of an electron-positron pair as result of absorption of gamma quantum in Coulomb field of nucleus:

Pair production through interaction of gamma quantum with Coulomb field of nucleus

Annihilation of $e^+ + e^-$ pair.

$\gamma \to e^+ + e^-$

Due to energy & momentum conservation, pair production requires minimum photon energy, threshold energy = $2m_e c^2 = 1.02\,\text{MeV}$

Absorption Coefficient

The absorption coefficient for pair production μ_{Pair} exhibits different energy dependencies in different regimes:

Change in intensity dI over path length dx proportional to incident intensity I & attenuation coefficient μ:

$$-dI = \mu \cdot I \cdot dx \tag{12}$$

Integral Form

Integrating the differential equation yields the exponential attenuation law:

$$I = I_0 \cdot e^{-\mu x} \tag{13}$$

where:

I_0 is initial intensity

I is intensity after traversing thickness x

μ is linear attenuation coefficient

Near threshold: $\mu_{\text{Pair}} \propto Z^2 \left(E_\gamma - 2m_e c^2\right)$

For high energies: $\mu_{\text{Pair}} \propto Z^2 \ln E_\gamma$

Pair production becomes dominant interaction mechanism for photons above approximately 10 MeV. The total absorption coefficient is sum of individual interaction coefficients:

$$\mu_{\text{tot}} = \mu_{\text{Photo}} + \mu_{\text{Compton}} + \mu_{\text{Pair}}$$

where:
μ_{Photo} is photoelectric absorption coefficient

μ_{Compton} is Compton scattering attenuation coefficient

μ_{Pair} is pair production absorption coefficient

When **high-energy gamma radiation** ($E_\gamma > 100\,\text{MeV}$) enters matter:

$$e^{\pm} \rightarrow \gamma + e^{\pm} \quad \text{(Bremsstrahlung)}$$

Pair production $\Leftrightarrow$ Bremsstrahlung

Electron-photon shower (electromagnetic cascade)

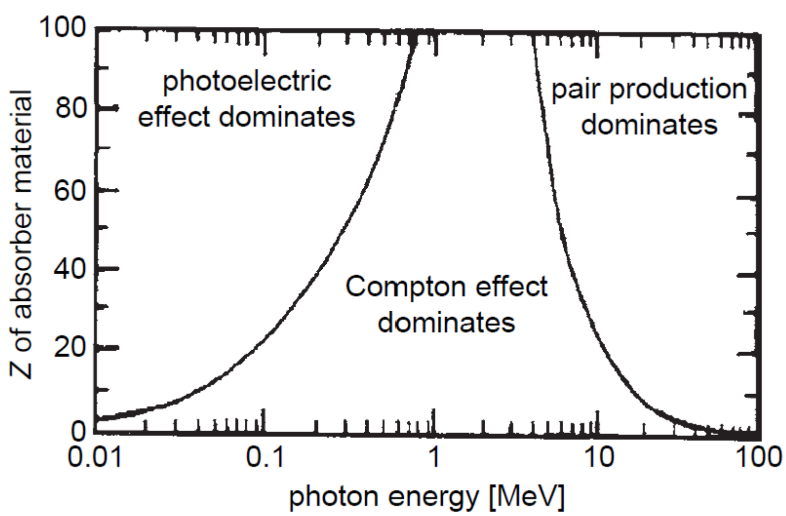

Figure 4: According to photon energy E_γ & atomic number Z classification of dominant interaction [9]

1.5 Experimental Setup

[10] **Scintillator**
We will use 2 organic plastic scintillator. The measurement scintillator counters remained unchanged. 1st, convert excitation of, e.g., crystal lattice caused by energy loss of particle into visible light & 2nd transfer this light via light guide to optical receiver (PMT). Scintillation efficiency, light output, emission spectrum & scintillator light decay time. It described as ratio of energy of emitted photons to total energy absorbed in scintillator. After charged-particle energy deposition plastic scintillator materials luminesce. In maximum-sensitivity wavelength range of PMT (400 nm) one or two (sometimes even three) fluorescent agents added

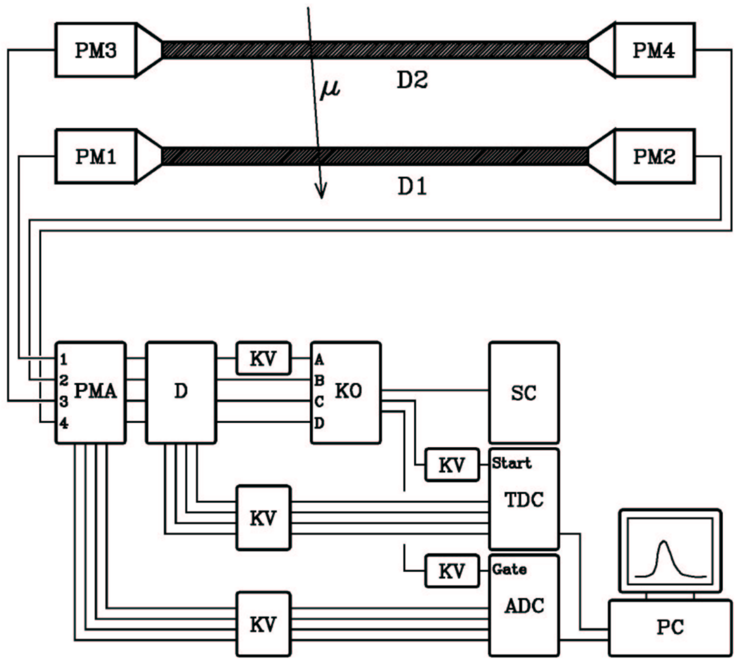

Figure 5: Schematic experimental set up [5]

to basic material acting as wavelength shifters for obtain light output. Excitation of molecules of polymer transferred to first fluorescent & de-excitation of fluorescent component produces light of longer wavelength

Light propagator (light guide) Shape of long bars , way of light collection is to use effect of internal (total) reflection for scintillator .

Photomultiplier tube (PMT)
For measurement of fast light signals we will use PMT. From scintillator – ejects electrons from photocathode via photoelectric effect for visible light or uv range. For negative high voltage measurements applied to photocathode & for positive high voltage measurements applied to anode. Photoelectrons focussed by electric field onto first dynode, which is part of multiplication system. The voltage between anode & photocathode subdivided by resistors in fig(6).

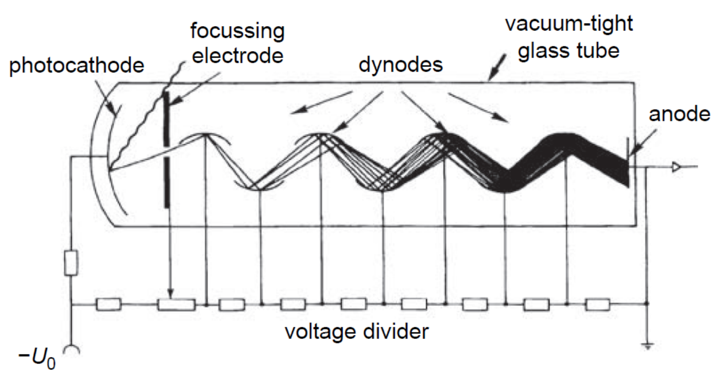

Figure 6: Diagram of PMT representing photocathode, dynodes, anode, & voltage divider.

Amplifier (PMA)
It amplify weak signals from detector & to drive through cable that connects preamplifier with rest of equipment. Since input signal is weak, preamplifiers normally mounted as close as possible to detector so as to minimize cable length.

Discriminator
Discriminator is analog-to-digital converter. It is device responds only to input signals with pulse height greater than certain threshold value. If it satisfied, discriminator responds by logic signal, if not, no response made. The value of threshold can adjusted by screw on front panel. From PMT use of discriminator for blocking out low amplitude pulses. Discriminator operation only signals whose amplitude greater than fixed threshold trigger an output signal [13].

Coincidence unit (KO) If two or more logic signals coincident in time & generates a logic signal if **true** & no signal if **false** then coincidence unit determines. Between two pulses made in number of ways the electronic determination of coincidence. One method to use transmission gate and other method used to sum two input pulses & to pass summed pulse through discriminator set at height just below sum of two pulses.

Analog to digital converter (ADC)
ADC device converts information contained in analog signal to digital form. We will use Wilkinson method for analog to digital conversion for spectroscopy ADC . Input signal first used to charge capacitor then it discharged at constant rate. At start of discharge, scaler counting pulses from constant frequency oscillator gated on. When capacitor completely discharged, scaler gated off. The contents of scaler is then number proportional to charge on capacitor.

Time to digital converter (TDC)
For measurement time interval n digital form, we use digitize TAC(Time-to-Amplitude Converter) using ADC. Start signal to gate on scaler, counts constant frequency oscillator. This scaler gated off to number proportional at arrival of second STOP signal, to time interval between pulses.

Scaler (SC) Scaler counts number of pulses fed into input. Scalers require shaped signal in order to function correctly; thus it necessary to have pulse shaper process signals from detector before they can be counted by scaler.

Cable delay (KV) The signals arrive after start signal from coincidence unit to TDC/ADC.

2 Evaluation

Properties of the signal & setting up electronics
In the first part of the experiment, the goal was to get to know the electronics by understanding the properties of incoming & outgoing signals of the components. The first step in performing the experiment is to analyze and verify its components. Furthermore, we will calibrate the detectors using cobalt-60 & sodium-22 source to be able to measure muons arrival time & energy.

Signal of PMT & Amplifier
We use the oscilloscope to simultaneously display amplified & non amplified PMT-signal start from signal output of PMT A & amplifier. Then we determine amplification factor from signal amplitudes in multiple amplitude ranges.
In Fig. 7, the yellow trace (channel 1) represents raw analog signal directly from the PMT. It is a relatively small negative pulse. The orange line measures minimum value (Channel 3) output from Amplifier (PMA). We have taken 16 oscilloscope images as measurements for the amplification factor. These measurements are shown in Table 1. The amplification factor is then fitted as a linear function in 8. The resulting amplification

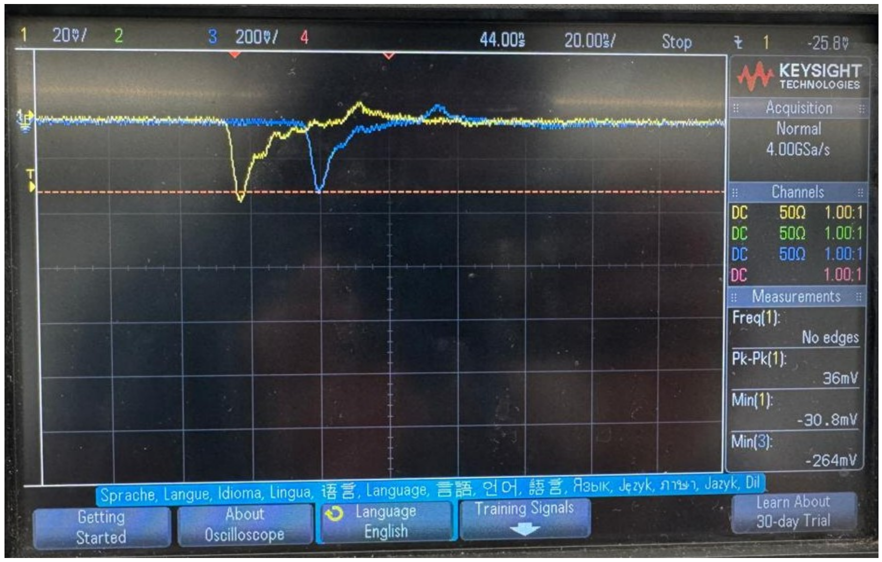

Figure 7: Oscilloscope showing PMT (yellow) with peak-to-peak amplitude & amplifier (blue) signals

factor is 6.7 ± 0.5.

Measurement	1	2	3	4	5	6	7	8
U_{PMT} [mV]	30.8	27.6	30.8	30.0	42.9	30.8	33.2	32.4
U_{Amp} [mV]	264	248	280	248	336	272	296	272
Measurement	9	10	11	12	13	14	15	16
U_{PMT} [mV]	47.7	27.6	46.1	38.1	36.5	29.2	30.0	29.2
U_{Amp} [mV]	408	248	348	312	296	256	264	248

Table 1: The measurements of the peak voltage of the PMT and the amplifier during a detection.

The amplifier is necessary because PMT signals are often too small to be reliably analyzed by subsequent electronic modules (like Discriminator or ADC) without getting lost in electronic noise.

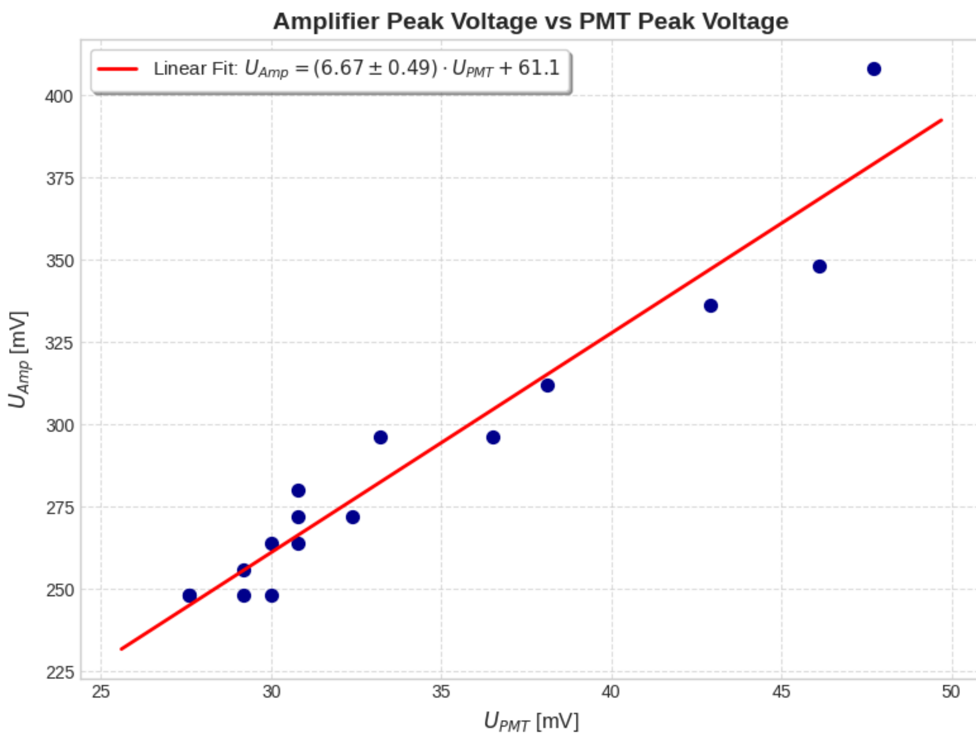

Figure 8: Amplifier U_{Amp} Maximum signal vs Maximum PMT signal U_{PMT}. The data points fitted with linear function $f(U_{\mathrm{PMT}}) = a \cdot U_{\mathrm{PMT}} + b$, where the slope a represents the amplification factor.

Start & Stop Signals

We will now prove that signals from PMT A & PMT B from the discriminator output are coincident. We measured this on oscilloscope in Fig. 9. We got coincident logical signals.

Figure 9: Oscilloscope measurement of discriminator of PMT A (yellow) & discriminator of PMT B in blue with a coincident signal.

Figure 10: Oscilloscope measurement of discriminator of PMT A (yellow) & discriminator of PMT B in blue. Here there was no two coincident signals.

Now we verified that the signal from the amplifier reached at the ADC within the gate window. This can be seen in the oscilloscope measurement in Fig. 11. They indeed overlap.

Figure 11: Measurement from PMT A of amplified signal (yellow) & gate signal (blue) of ADC.

Now the verification of TDC start signal from coincidence & stop signal from discriminator is done. The stop-signal is supposed to reach after the start signal. An oscilloscope measurement that indeed verifies this is shown in Fig. 12.

Figure 12: Oscilloscope measurement of start-signal from coincidence unit (yellow) & stop-signal from discriminator (blue).

Impact of background light

We will test background light influences on the PMTs. We measure with & without room light by using ADC spectra in coincidence condition A. The measurement time was 180 seconds for both. The number of counts in PMT A are then determined.When the light was on, the number of events were higher & when the light was off the number of events was less.

Detection rates for both conditions:

$$R = \frac{N}{t} \tag{14}$$

where: N is total number of events (counts), t is measurement time (in seconds) & R average event rate (in Hz)

Extracting ADC data for PMT A for light comparison with results in Tab. 2.

	Rate [1/s]	Total counts
light ON	160	28744
light OFF	154	27808

Table 2: Rate and total counts for the ADC spectrum without background light and with background light.

This means that the background light has very small effect on the PMTs.

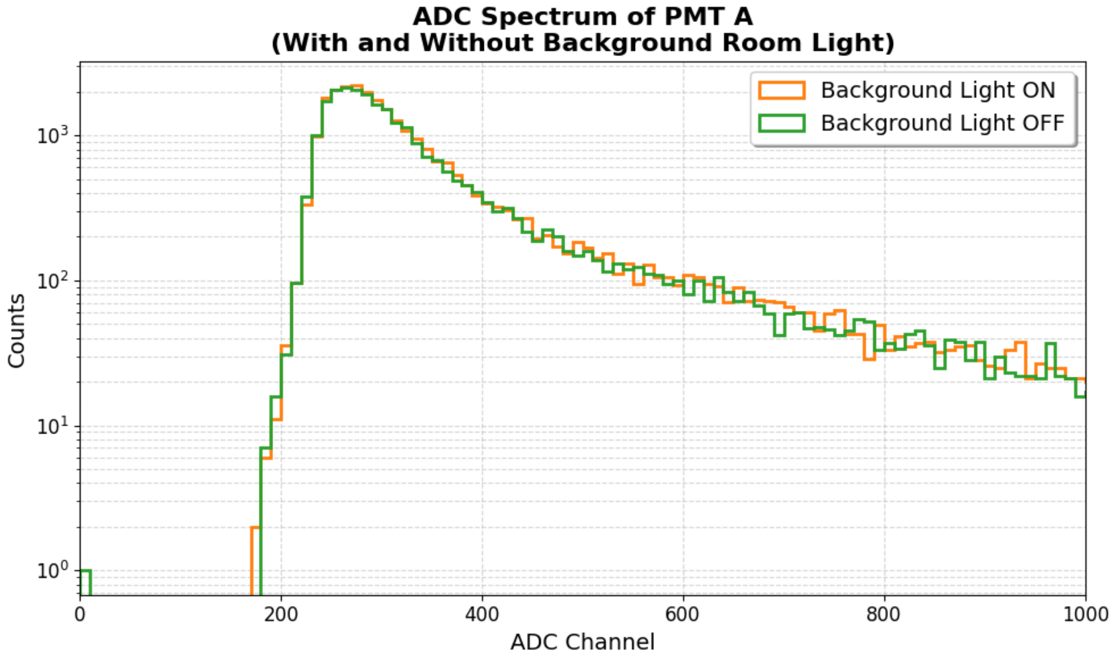

Figure 13: ADC spectra with & without light at PMT A

Here, we can see in figure 13 the two spectra overlay almost perfectly but Light ON (orange) will sit slightly higher than Light OFF (green) on Y-axis.
This graph verifies that the background of counts we calculated in the beginning are not high-energy CRs, they are primarily low-energy noise (photons from room lights leaking through tiny gaps in black tape & triggering single photoelectrons at photocathode).

Time Calibration
In this part we established relation between channel number & time delay at TDC by using PMT (A,B,C,D). We set up 2, 4, 8, 16, 32 (in ns) time delay for each PMT using cable delays with their respective coincidence condition. Then we recorded spectra & found the channel number where peak number of events occur. This is shown for PMT A in figure 14.

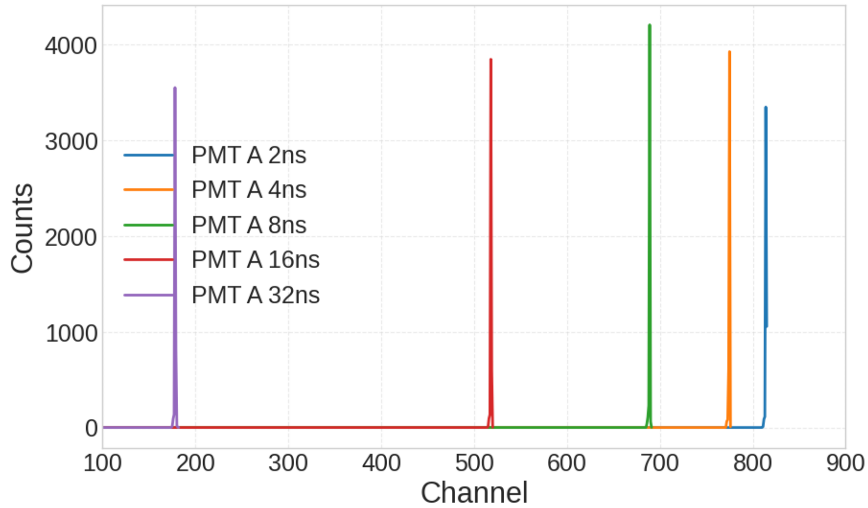

Figure 14: For PMT A TDC spectra in different cable delays of 2ns, 4ns, 8ns, 16ns & 32ns.

Now we describe linear (time calibration) function. Here a linear function is fitted to the peak channels & inserted time delays. This is shown in Figure 15.

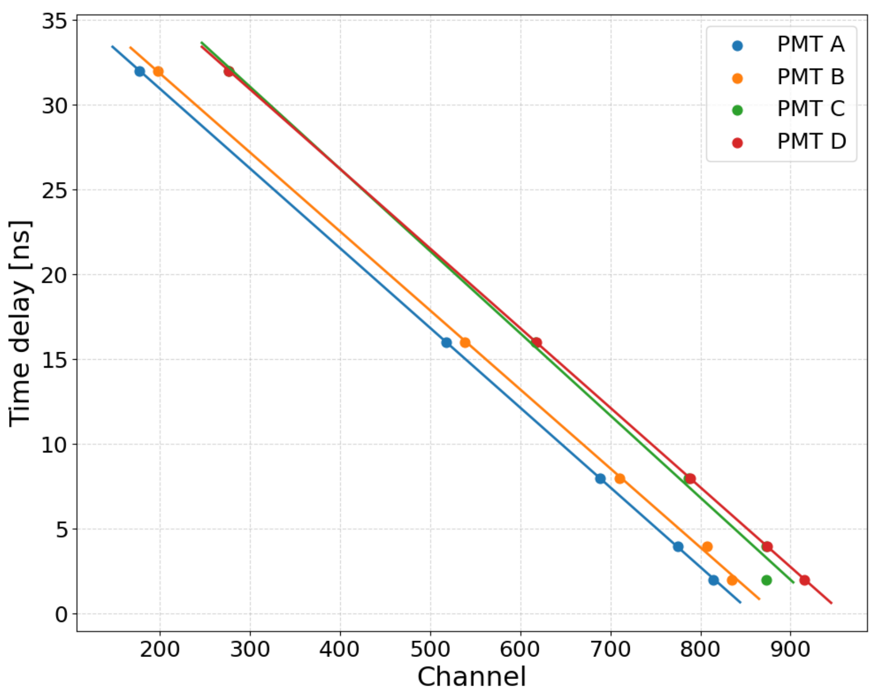

Figure 15: Peak channel number linear fit all PMT vs time delay. Here x-axis (Channel) is digital number output by TDC where the peak was measured & Y-axis is time delay (in nanoseconds).

Time calibration functions:

$$\begin{aligned}
\text{PMT A}: f(\text{channel}; a, b) &= -49 \pm 4\,\text{ps/a.u.} \cdot \text{channel} + 41 \pm 3\,\text{ns} \\
\text{PMT B}: f(\text{channel}; a, b) &= -47.1 \pm 0.2\,\text{ps/a.u.} \cdot \text{channel} + 41.35 \pm 0.08\,\text{ns} \\
\text{PMT C}: f(\text{channel}; a, b) &= -48.1 \pm 1.6\,\text{ps/a.u.} \cdot \text{channel} + 45.4 \pm 1.2\,\text{ns} \\
\text{PMT D}: f(\text{channel}; a, b) &= -47.0 \pm 0.2\,\text{ps/a.u.} \cdot \text{channel} + 45.05 \pm 0.08\,\text{ns}
\end{aligned}$$

The slope (a) of these lines represents resolution of TDC. PMT(A,B,C,D) are plugged into exact same TDC. The graph verify TDC treating time scaling similarly for all PMTs.

Energy Calibration

In this part the muon energy loss at ADC channel numbers is calibrated to deposited energy in the scintillator. We will keep the radioactive source ^{60}Co in the center between the two scintillators & measure 3 min for ADC spectra for PMT(A,B,C,D) under each coincidence condition. Then we do the same for ^{22}Na. We have three points: zero-energy (peak), Compton edges of ^{60}Co & ^{22}Na from gamma emissions. These points are used for calibration by linear fitting. We already calculated Compton edges in the equations 6 & 7 for ^{22}Na & ^{60}Co. For example we got zero-point energy for PMT A with ADC measurement under coincidence condition C. Then we measured the zero-point energy for all PMTs(A,B,C,D). Then we measure with ^{60}Co & ^{22}Na source for all PMTs. After that we measure without a source for all PMTs.

To determine the exact channel corresponding to zero deposited energy for PMT C a Gaussian function is fitted which takes the form:

$$f(x) = A \cdot \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right) \tag{15}$$

where A describes amplitude (peak height), μ is mean channel, & σ is standard deviation (peak width).

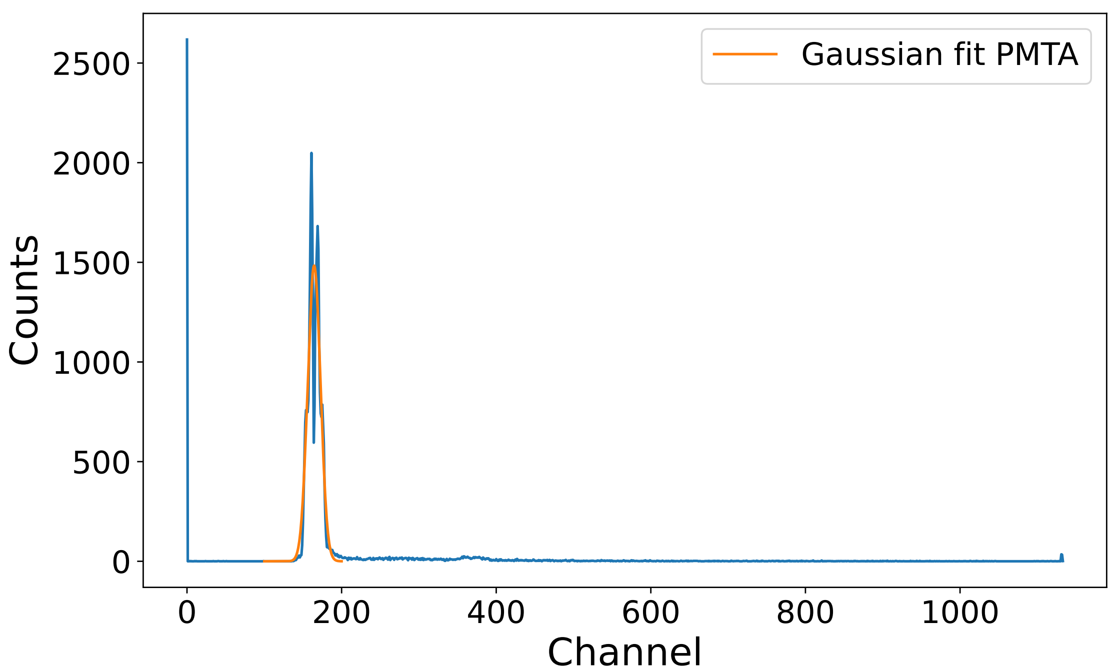

Figure 16: This shows Zero-peak energy & fit for PMT A with ^{60}Co source under coincidence condition C in blue & Gaussian fit in orange.

Moyal distribution fit (approximation of Landau distribution) is used for determining the Compton edge for energy calibration with & without source [15].

The Moyal fit function takes the form:

$$f(x) = A \cdot \exp\left(-\frac{1}{2}\left(2.22 \cdot \frac{x-\mu}{\sigma} + \exp\left(-2.22 \cdot \frac{x-\mu}{\sigma}\right)\right)\right) \tag{16}$$

where A represents amplitude, μ dictates peak location, & σ describes scale width. Calculation of Moyal function for Compton edge analysis is done with the Full Width at Half Maximum (FWHM). For PMT A this is shown in figure 17. The FWHM for Moyal distribution is related to the width parameter:

$$\text{FWHM} = 1.6175 \times \sigma[15] \tag{17}$$

This constant factor (1.6175) is property of Moyal distribution shape, representing ratio between full width at half maximum & scale parameter σ.

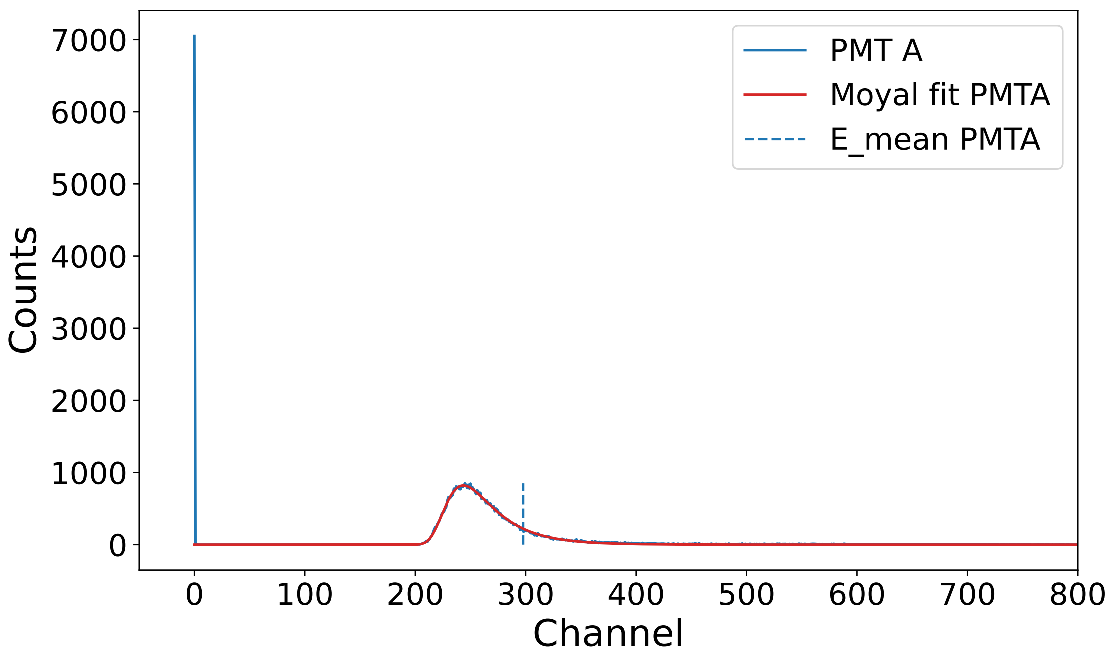

Figure 17: ADC(X-axis Channel) spectrum for PMT A with source ^{60}Co

The graphs Fig. 16 and Fig. 17 display complete energy calibration measurement for PMT A with source ^{60}Co. On the left the Channels 0–100 show the zero-energy peak & the right Channels 200–400 show the Compton continuum. The red line is the fitted Moyal function & the dashed blue line ($E_{\text{mean PMT A}}$) is the Compton edge.
Moyal distribution (red curve) was fitted to the continuum. Both the zero-energy peak x_0 energy & Compton edge x_C energy is known and the corresponding channel numbers were determined.
Compton Edge Position:

$$x_C = \mu + 1.6175\sigma$$

Energy calibration linear fit function:

$$E(x) = a \cdot x + b$$

where $a = \frac{E_C}{x_C - x_0}$ & $b = -a \cdot x_0$.

For each PMT, calibration parameters (slope a & intercept b) were calculated for ^{60}Co & ^{22}Na radioactive sources then averaged to obtain final calibration function below:

PMT	A	B	C	D
a [keV/a.u.]	9.43 ± 0.05	8.32 ± 0.02	8.54 ± 0.02	37.0 ± 0.2
b [keV]	-1525 ± 8	-521.8 ± 1.4	-909.3 ± 2.6	-5360 ± 26

Table 3: Energy calibration function parameters for each PMT

Overlap Coincidence

We will verify same timing of overlap coincidence with radioactive source ^{60}Co by using PMT A & PMT B with TDC spectra under coincidence condition A. Then we measure with ^{60}Co using PMT A+B with cable delay 16 ns & without. Under coincidence condition A: When PMT A detects a signal events are recorded & under coincidence condition A+B: When both PMT A & PMT B detect signals simultaneously, events are recorded. However the plots show that there are fewer events than under coincidence condition A. When both PMT A & PMT B detected signals simultaneously they are within coincidence. 2D TDC correlation histograms are shown in 18 , where plot x-axis is TDC A and y-axis is TDC B. For coincidence condition A and no additional time delay on PMT A, a narrow line can be seen for PMT A & TDC B shows a broad distribution line at PMT B. This is because A always starts the measurement process. For coincidence (A+B) with no time delay, then both PMTs can start the measurement process and a corner is visible in the 2D TDC spectrum. For coincidence (A+B) with 16 ns delay on PMT A the corner disappears because now PMT A again determines when a signal is measured.

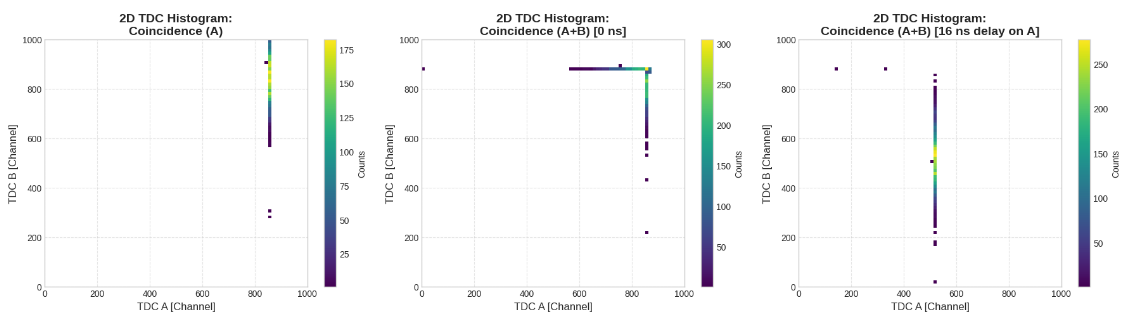

Figure 18: 2D TDC correlation spectra of PMT A & PMT B.

Figure 19 shows 1-dimensional TDC spectra for both PMT A & B under coincidence conditions. In the top row the system is under condition A, where PMT A (top left) is sharp peak, while PMT B (top right) shows broad time distribution. The bottom row presents spectra under condition A+B after 16 ns cable delay was applied to PMT A. In this configuration, PMT A (bottom left) has same sharp peak shifted to a lower channel, & PMT B (bottom right) continues to show a broad, varying time distribution.

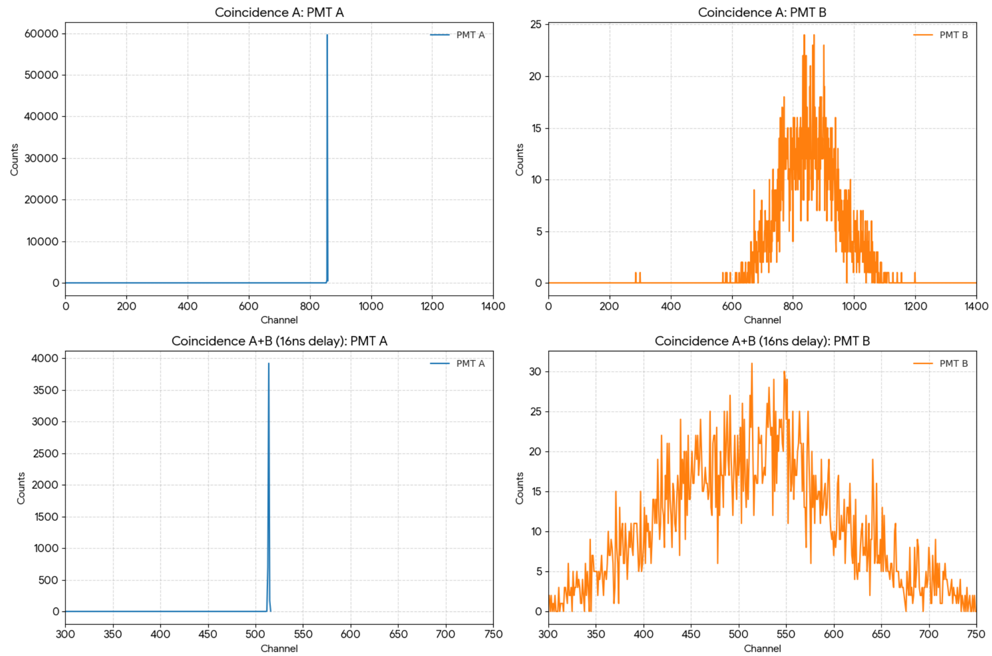

Figure 19: In the left column TDC spectra are shown for PMT A for coincidence condition A with no additional time delay on PMT A & coincidence condition A+B with 16 ns time delay. The right column shows the TDC spectra for PMT B under the same conditions.

Detector performance with Cobalt
Now we will get additional inquiry about the detector by using the ^{60}Co-source.
Attenuation in scintillator
In this part we have taken seven measurements with PMT A & B. First we measure 25cm to left & right from center with PMT A & B. Then we keep source ^{60}Co at center using PMT A & B.
In Fig. 20 the orange curve shows peak height. The blue curve shows second peak height and the green curve shows third peak height which are all similar to a Landau distribution. ^{60}Co is placed at three positions along scintillator center (d = 0 cm from center), 25 cm to the right (closer to PMT A) & 25 cm to left (farther from PMT A).
As the source ^{60}Co moves away from PMT A, which is at the right, scintillation light travels a longer distance through the plastic scintillator before reaching PMT A. During this journey some photons are absorbed by scintillator molecules, some photons scatter out of the direct path & some photons escape through scintillator surface.
Fit an exponential decay function to the total number of counts at each distance:

$$f(d; a, b, c) = a \cdot \exp(-\frac{d}{b}) + c$$

In 22 the exponential decay of total number of counts over distance is fitted and the count values are shown in Tab. 4. The fitted attenuation length is $b = 60$ cm, which leads to a attenuation coefficient of $\mu = 1/b = 0.0167$ cm^{-1}.

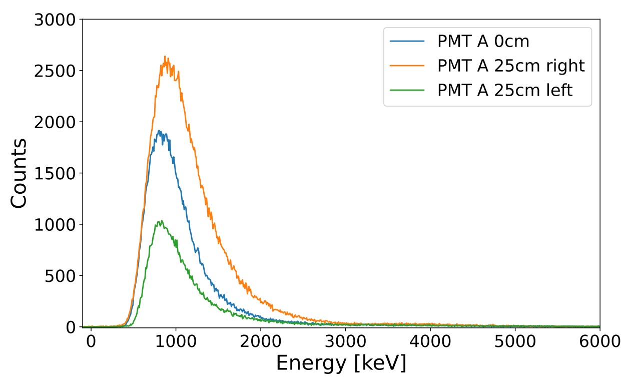

Figure 20: ADC spectra for PMT A when the source is in the center, 25 cm to the right and 25 cm to the left.

The peak positions for the ADC spectra in Fig. 20 shifts slightly. When less light reaches at PMT then measured energy appears lower due to reduced photoelectron statistics. Fig. 21 shows the ADC spectra for PMT B which as expected shows the opposite total number of counts with distance trend to PMT A.

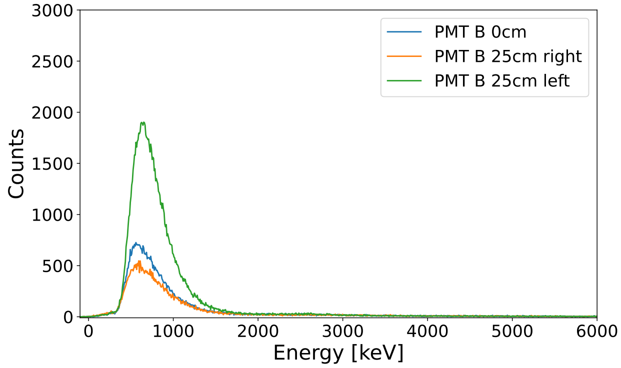

Figure 21: ADC spectra for PMT B when the source is in the center, 25 cm to the right and 25 cm to the left.

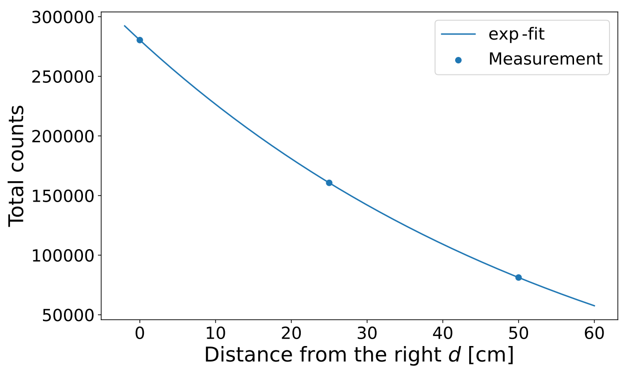

Figure 22: Exponential decay fit to the total number of counts at each of the three source positions.

^{60}Co source position	Total counts
25 cm right	280388
center	160650
25 cm left	81255

Table 4: Total number of counts at the three source positions.

Free & coincident energy spectra

In this step we measure 7 minute comparison of free & coincident energy spectra with radioactive source ^{60}Co by using PMT A & B. Two ADC energy spectra measurements are recorded with ^{60}Co source at detector center.

Here the blue curve shows the ADC spectrum for PMT A under condition A (Free Spectrum) which shows the highest peak height compared to the coincident spectrum. The orange curve shows PMT A under Condition A+B (Coincident Spectrum, scaled), & the green curve shows PMT B under Condition A+B (Coincident Spectrum, scaled) in Fig. 23.

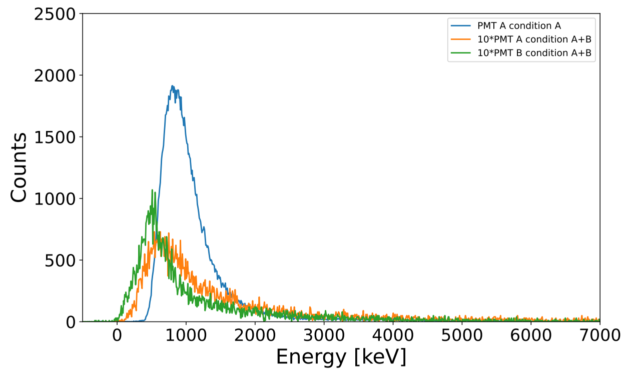

Figure 23: ADC spectra for PMT A under coincidence condition A and ADC spectra for PMT A and B under coincidence condition A+B.

Speed of light

In this part the speed of light in the scintillator medium is measured using ^{60}Co source placed at three distinct locations: the center of detector, 25 cm to right (closer to PMT A) & 25 cm to left (closer to PMT B). For each position, time difference between signals arriving at PMT A & PMT B was recorded via TDC spectra. In Fig. 24, Gaussian functions fitted tTDC distributions to extract central time delay (μ-parameter) for each source position. Those parameters are

$$\mu_{\text{right}} = 16.68 \pm 0.09\,\text{ns}$$
$$\mu_{\text{center}} = 17.03 \pm 0.06\,\text{ns}$$
$$\mu_{\text{left}} = 18.91 \pm 0.05\,\text{ns}$$

Fig. 25 plots the extracted peak time delays against the source displacement distance. The data is modeled with a linear fit, where the inverse of the resulting slope represents the effective speed of light in the medium. The effective speed of light in medium is $c_{\text{eff}} = (1.90 \pm 0.07) \cdot 10^8$ m/s with a refractive index $n = 1.57 \pm 0.06$. This is around the actual refractive index for the material of $n = 1.58$.

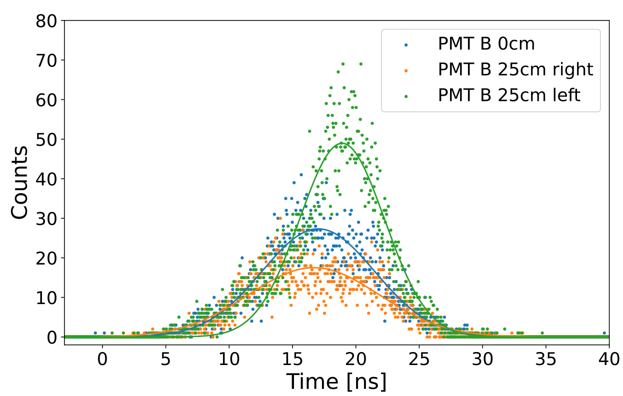

Figure 24: TDC spectra for PMT B for the three source positions with Gaussian fits.

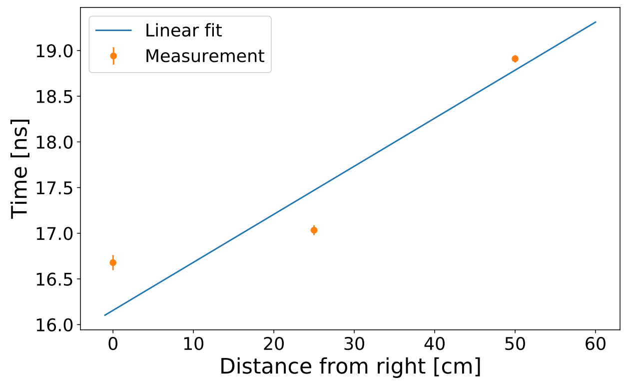

Figure 25: Linear fit to the fitted mean positions of the Gaussians.

Properties of atmospheric muons
Next the properties of atmospheric muons are looked at.
Detector resolution
We will measure at TDC spectrum of time differences between PMTs was recorded with cosmic ray particles. The measurement used coincidence condition A+C, so events where a particle goes through both plates counted.
The resulting time-difference spectrum for PMT C is shown in Fig. 26. Fitted Gaussian function distribution:

$$f(t; A, \mu, \sigma) = A \cdot \exp\left(-\frac{1}{2} \cdot \frac{(t-\mu)^2}{\sigma^2}\right) \tag{18}$$

The temporal resolution (τ) defined by the Full Width at Half Maximum (FWHM) of Gaussian fit :

$$\tau = 11.79 \pm 0.12\,\mathrm{ns} \tag{19}$$

The central time dela (χ), means uncertainty of muon, calculated by temporal resolution by speed of light within scintillator medium (c_{eff}):

$$\chi = c_{\mathrm{eff}} \cdot \tau = 224 \pm 9\,\mathrm{cm} \tag{20}$$

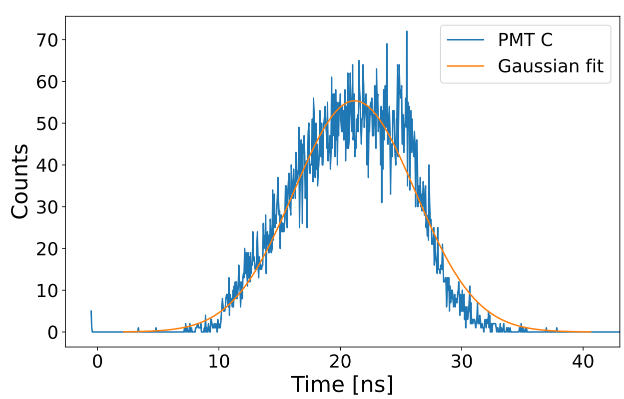

Figure 26: TDC spectrum for PMT C under coincidence condition A+C with a Gaussian fit for the temporal resolution.

In Fig. 26 the x-axis is the time recorded by the TDC & y-axis is the counts. This shows how many times a cosmic muon produced that specific time difference between PMT A and PMT C.
Preparation of detector
Energy spectra were recorded using ADCs. The measurement (15 min) was taken under coincidence condition

A+C, meaning an event was only recorded if a particle went through both the upper & lower scintillator plates.
Fig. 27 shows resulting energy spectra for both PMT A (blue) and PMT C (orange). The primary shape of both curves is asymmetric peak characterized by initial vertical & long tail extending toward higher energies. This specific shape is Landau distribution for energy loss Minimum Ionizing Particles like muons.

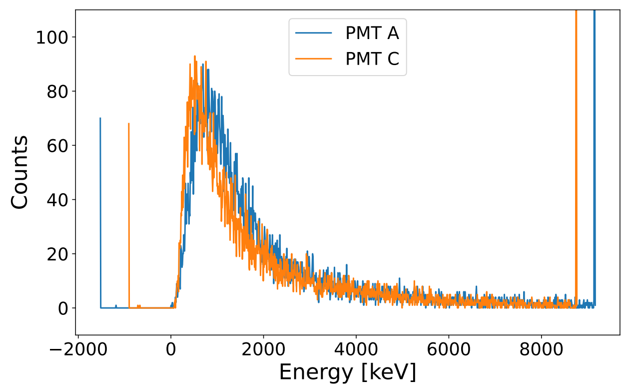

Figure 27: ADC spectra for PMT A and C under coincidence condition A+C.

Fig. 28 shows the 2D correlation map for ADC spectra from PMT A and PMT C. It shows that the main signal in the lower left is from single-particle events. The spread to the upper right can happen when a muon was detected at different positions in the scintillators relative to PMT A and PMT C.

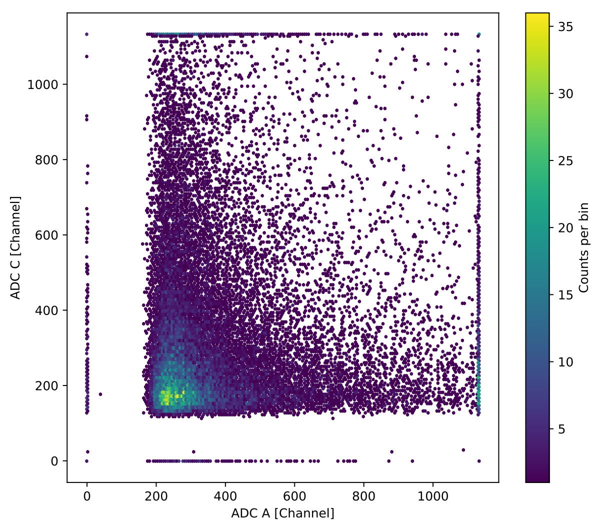

Figure 28: 2D ADC correlation histogram for PMT A and PMT C.

Fig. 29 shows the correlation between ADC and TDC for PMT C. The main signal is in the column in the left of the plot. Here most of the particles deposit their energy as it is the Landau distribution peak. The spread of events to the right shows a correlation between pulse height and timing which could come from time walk for the PMT.

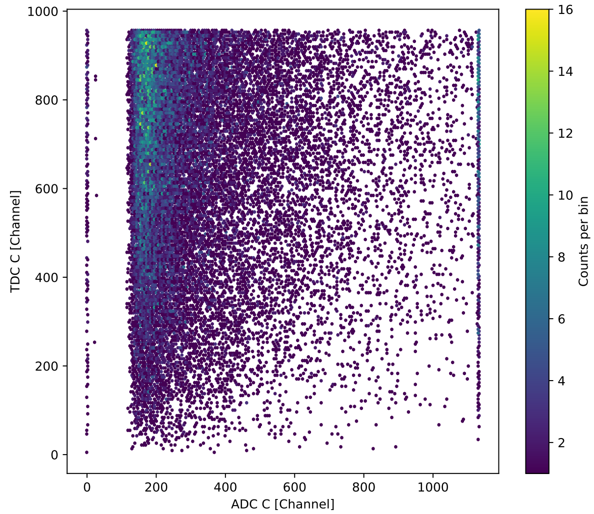

Figure 29: Correlation histogram for the TDC spectrum of PMT C and the ADC spectrum of PMT C.

Intensity & energy loss

To determine the energy loss of the cosmic muons the detectors are crossed and the raw ADC spectra for PMT A & PMT C are fitted using Moyal distribution from Eq. 16.Fig. 30 shows results of these fits on ADC spectra of PMT A and PMT C. The Moyal function fits asymmetric shape of energy peak.

The vertical dashed lines in fig 30 shows calculated mean deposited energy (E_{mean}) for each detector. It is important difference between the Most Probable Value (peak distribution) & mean energy. The mean energy E_{mean} is 1749 ± 11 keV for PMT A and 1401 ± 5 keV for PMT C. This is slightly lower than the calculated energy loss of 3780 keV from preparation Eq. 4.

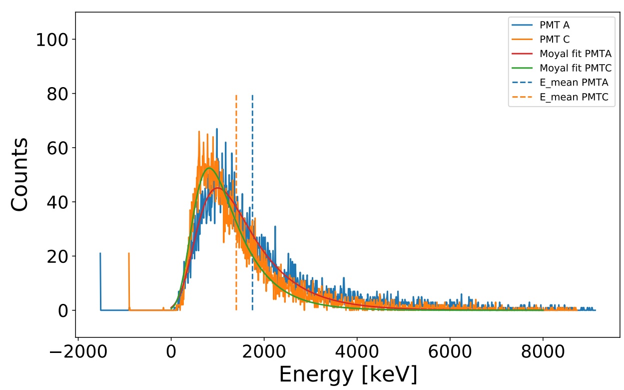

Figure 30: ADC spectra for PMT A and C with coincidence condition A+C with fitted Moyal function and Compton-mean energies.

Now the intensity is calculated. The count intensity I_{c} comes from the total number of counts divided by the area $A = (32\ \mathrm{cm})^2$ and the measurement time $t = 15 \cdot 60$ s.

$$I_{\mathrm{c}} = \frac{N}{A \cdot t} = \frac{8933}{(32\,\mathrm{cm})^2 \cdot 900\,\mathrm{s}} = 97\ \frac{\mathrm{counts}}{\mathrm{m}^2\,\mathrm{s}}$$

The energy flux F_{e} for each PMT is calculated by summing over each energy bin E_{i} weighted with their count values c_i and dividing by the area A.

$$F_{\mathrm{e,PMTA}} = \frac{1}{A}\sum_i c_\mathrm{i} E_\mathrm{i} = 160.9\ \mathrm{GeV\,m^{-2}}$$
$$F_{\mathrm{e,PMTC}} = \frac{1}{A}\sum_i c_\mathrm{i} E_\mathrm{i} = 131.2\ \mathrm{GeV\,m^{-2}}$$

Electromagnetic & muonic component

Now we try to get rid of the electromagnetic part by putting a table with lead bricks between the two detector plates. The idea is that the lead should stop most of the electromagnetic stuff before it reaches the lower detector.

To check this we do another measurement for 15 minutes with the lead in place and compare the ADC spectra of PMT A and PMT C to the ones from Fig. 30, where there was no lead.

The ADC spectra with and without lead are shown in Fig. 31 for PMT A and Fig. 32 for PMT C. One can see that the mean energy E_mean stays about the same for PMT A and increases for PMT C when the lead is there.

This is probably because the electromagnetic part is being removed. The results comparing the spectra with and without the lead layer are shown in Tab. 5. The total number of events N gets smaller for PMT A when the lead is between the detectors since some of the electromagnetic part is filtered out.

The deposited energy which is calculated by adding up the energy bins times the counts also becomes smaller for PMT A when the lead is used but higher for PMT C. This could be because of shift of energy peak to higher energy from extra secondary particles produced in upper scintillator that then are detected in PMT C and we only count events where PMT A+C happen at the same time.

Still most of the events are detected since they mainly come from the muon part of the cosmic rays.

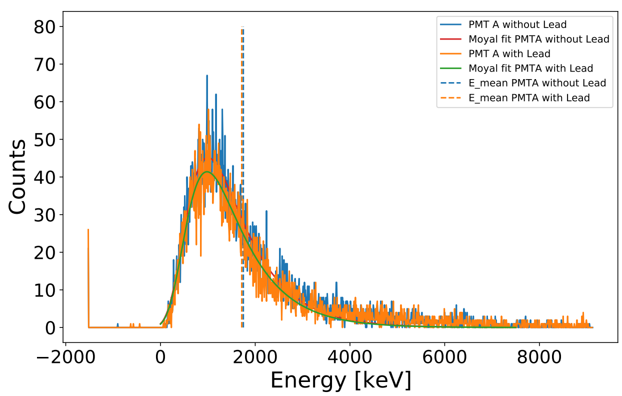

Figure 31: ADC spectra of PMT A with coincidence condition A+C with and without a layer of lead placed between the detectors as well as Moyal fits and Compton-energies.

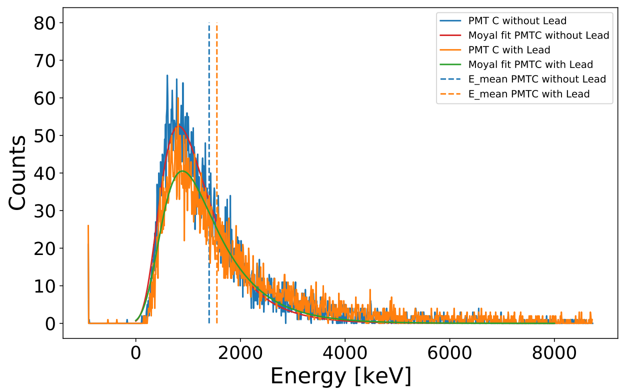

Figure 32: ADC spectra of PMT C with coincidence condition A+C with and without a layer of lead placed between the detectors as well as Moyal fits and Compton-energies.

PMT	lead	$E_{\text{deposited}}$ [GeV]	N
PMT A	without lead	16.5	8933
	with lead	15.3	8140
PMT C	without lead	13.4	8933
	with lead	14.5	8140

Table 5: Deposited energy and number of counts for PMT A and PMT C with and without lead.

Speed & zenith angle dependence of muons

Take away the lead again and try to measure how fast the muons are. For this the top detector is rotated so the distance between the two plates is about $d = 1.20 \pm 0.01\,\text{m}$.

First do a TDC measurement with PMT A and C for 15 minutes with the coincidence A+C. This is done once at 0° and once at 180°. We do this twice because the TDC always has some offset. If we do two measurements the offset is the same and the muon flight time is basically half of the difference between the two.

The two TDC C measurements with Gaussian fits are shown in Fig. 33. The measured times are just the μ values from the fits:

$$\mu_{0^\circ} = 16.94 \pm 0.12 \text{ ns}, \mu_{180^\circ} = 25.37 \pm 0.10 \text{ ns}.$$

From this we get the speed:

$$\beta = \frac{v}{c} = \frac{d}{0.5 \cdot (\mu_{180^\circ} - \mu_{0^\circ})\, c} = 0.95 \pm 0.04.$$

So the muons are almost at the speed of light.

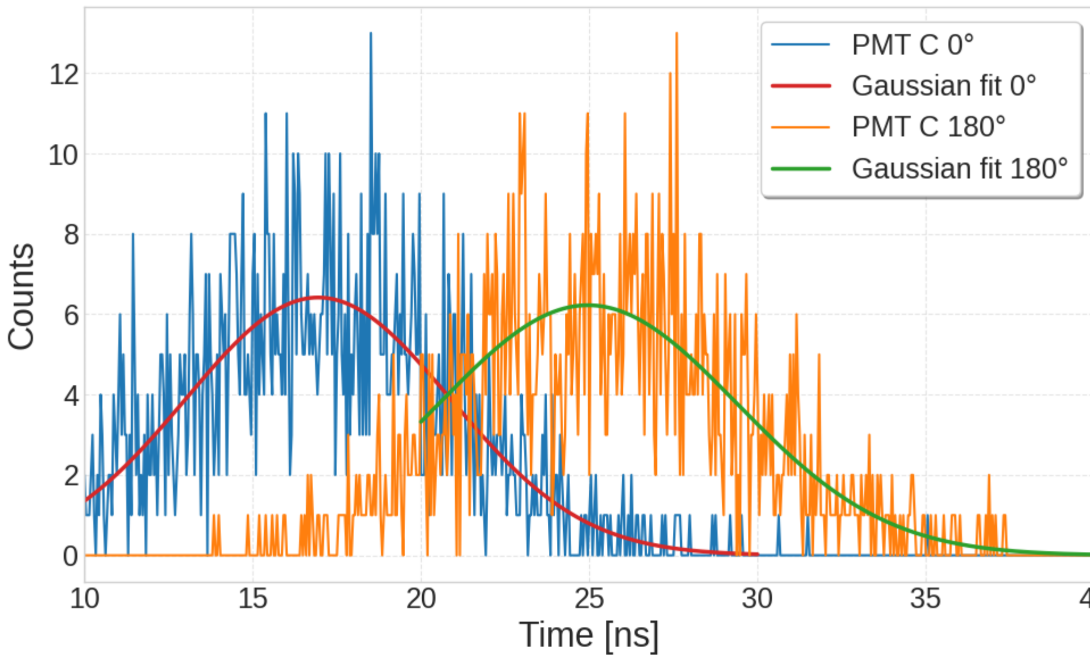

Figure 33: TDC spectra of PMT C for coincidence condition A+C for a rotation angle of 0° and 180° as well as Gaussian fits to the TDC spectra.

Next we look at how many events N we detect with the coincidence A+C for different rotation angles θ. For each angle we again measure for 15 minutes. The measured values can be found in Tab. 6.
We then fit the dependence on θ with the function

$$N(\theta) = a \cdot \cos^2(\theta) + b.$$

The result together with the fit is shown in Fig. 34. This shows that the muons kind of follow a $\cos^2(\theta)$ dependence.

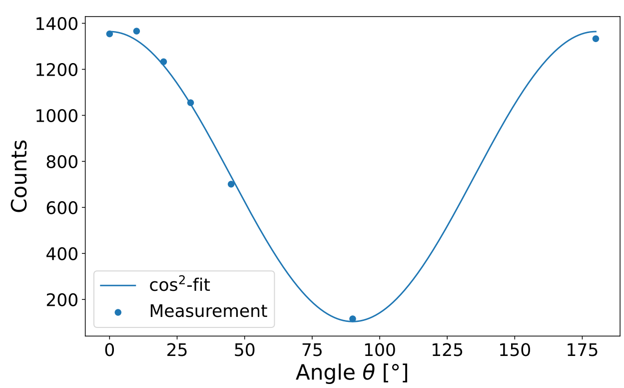

Figure 34: Total number of counts plotted for the different rotation angle as well as a $\cos^2$-fit to them.

θ [°]	0	10	20	30	45	90	180
N	1354	1366	1233	1055	701	116	1333

Table 6: Total number of counts N for different rotation angles θ. The measurement at 180° is physically equivalent to the measurement at 0°. Their small difference in counts is due to measurement statistics.

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